

Central Limit Theorem

It states that the distribution of sample means approaches a normal distribution (a bell curve) as a sample size increases, regardless of the population's original distribution shape.

Definition - Even if a population is skewed or non-normal, the average of multiple random samples taken from that population will tend to form a normal distribution.

Formula

The theorem relies on the transformation of the sample mean ($\bar{x}$) to standard normal distribution (z) using formula.

$$Z = \frac{\bar{x} - \mu}{\sigma \sqrt{n}}$$

$\bar{x}$ = Sample mean

μ = Population mean

σ = Population std deviation

n = sample size

Condition

sample must be independent, random and representative of population.

While $n \geq 30$ is a common thumb rule for a large enough sample.